Solutions to

Subtitle or Series

Main Title of the Book

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Chapter 1.

Basic Problems

Problem 1.

Solve the equation $x^2 - 5x + 6 = 0$.

Solution: We look for two numbers whose sum is 5 and whose product is 6: they are 2 and 3, so the polynomial factors as

$$x^2 - 5x + 6 = (x - 2)(x - 3). \tag{1.1}$$

A product is zero only when one of its factors is zero, so $x - 2 = 0$ or $x - 3 = 0$. As a check, the quadratic formula gives the same roots:

$$x = \frac{5 \pm \sqrt{5^2 - 4 \cdot 6}}{2} = \frac{5 \pm 1}{2}.$$

Answer to Problem 1.

$x = 2$ or $x = 3$.

Problem 2.

Let $S_n = 1 + 3 + 5 + \cdots + (2n-1)$ be the sum of the first n odd numbers.

(a)

Compute S_1, S_2, S_3 and S_4, and guess a formula for S_n.

Solution: Adding one odd number at a time,

$$S_1 = 1, \qquad S_2 = 1 + 3 = 4, \qquad S_3 = 4 + 5 = 9, \qquad S_4 = 9 + 7 = 16.$$

These are the squares of 1, 2, 3 and 4.

Answer to Problem 2 (a).

$S_n = n^2$.

(b)

Prove the formula of part (a) by induction.

Solution: For $n = 1$, $S_1 = 1 = 1^2$. If $S_n = n^2$ for some $n \geq 1$, then

$$S_{n+1} = S_n + (2n+1) = n^2 + 2n + 1 = (n+1)^2, \tag{1.2}$$

so the formula also holds for $n+1$, and by induction for every $n \geq 1$. This part is a proof, so it has no answer box.

(c)

Evaluate S_{10}.

Answer to Problem 2 (c).

$S_{10} = 10^2 = 100$ (no solution is needed, so only the answer is given).

(d)

How many consecutive odd numbers, starting from 1, must be added for the sum to exceed 1000?

Solution: We need the smallest n with $n^2 > 1000$, that is $n > \sqrt{1000} \approx 31.6$. Indeed $31^2 = 961 \leq 1000$ while $32^2 = 1024 > 1000$.

Answer to Problem 2 (d).

32 odd numbers.

Problem 3.

A right triangle has legs $a = 3$ and $b = 4$, as shown in Fig. 1.1. Find the length of the hypotenuse c and the angle θ opposite the leg a.

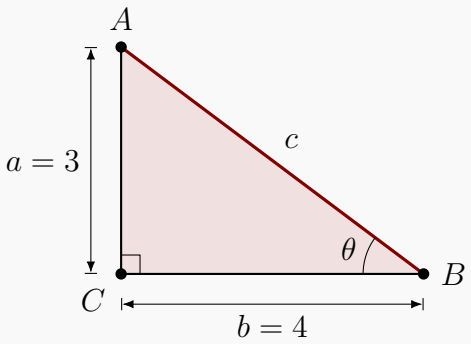


Figure 1.1: Right triangle with legs a and b.

Solution: By the Pythagorean theorem,

$$c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5.$$

The leg a is opposite θ and the leg b is adjacent to it, so

$$\tan\theta = \frac{a}{b} = \frac{3}{4}, \qquad \theta = \arctan\frac{3}{4} \approx 36.87^\circ.$$

Answer to Problem 3.

$c = 5$ and $\theta = \arctan(3/4) \approx 36.87^\circ$.

Chapter 2.

Figures and References

Problem 1.

Consider the function $f(x) = \frac{1}{4}x^2 + \frac{1}{2}$, shown in Fig. 2.1, and the point $P = (3, f(3))$ on its graph.

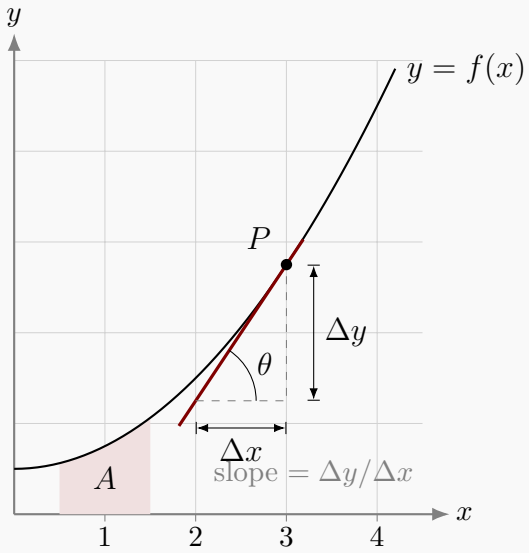


Figure 2.1: Graph of f, its tangent at P and the area A.

(a)

Find the slope of the tangent at P and the angle θ that it makes with the x axis.

Solution: The derivative is

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{x}{2},$$

so the slope at $x = 3$ is $f'(3) = 3/2$. In the slope triangle of Fig. 2.1, $\Delta y/\Delta x = \tan\theta$, hence $\theta = \arctan(3/2)$.

Answer to Problem 1 (a).

Slope $3/2$, and $\theta = \arctan(3/2) \approx 56.31°$.

(b)

Find the equation of the tangent at P.

Solution: Since $f(3) = \frac{9}{4} + \frac{1}{2} = \frac{11}{4}$, the line through P with slope $3/2$ is

$$y = \frac{11}{4} + \frac{3}{2}(x - 3) = \frac{3}{2}x - \frac{7}{4}.$$

Answer to Problem 1 (b).

$y = \frac{3}{2}x - \frac{7}{4}$.

(c)

Find the area A under the graph of f between $x = \frac{1}{2}$ and $x = \frac{3}{2}$.

Solution:

$$A = \int_{1/2}^{3/2} \left(\frac{x^2}{4} + \frac{1}{2} \right) \mathrm{d}x = \left[\frac{x^3}{12} + \frac{x}{2} \right]_{1/2}^{3/2} = \frac{27-1}{96} + \frac{1}{2} = \frac{13}{48} + \frac{24}{48}.$$

Answer to Problem 1 (c).

$A = \frac{37}{48} \approx 0.771.$

Problem 2.

Use Euclid's algorithm (Fig. 2.2) to find the greatest common divisor of 84 and 36.

Solution:

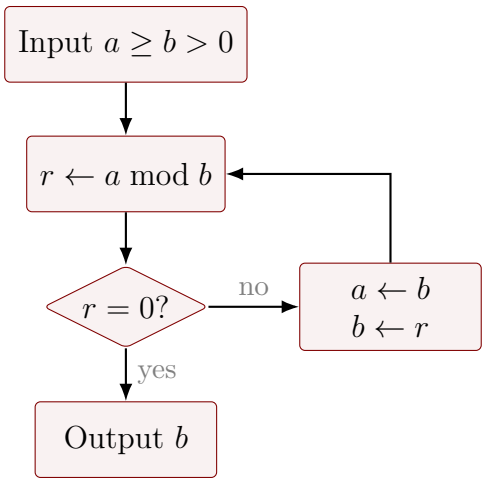


Figure 2.2: Euclid's algorithm.

Starting from $a = 84$ and $b = 36$, each step replaces (a, b) by $(b, a \bmod b)$:

Step	a	b	$r = a \bmod b$
1	84	36	12
2	36	12	0

The remainder is zero at step 2, so the algorithm outputs $b = 12$. Indeed $84 = 7 \cdot 12$ and $36 = 3 \cdot 12$, with $\gcd(7, 3) = 1$.

Answer to Problem 2.

$\gcd(84, 36) = 12$.

Problem 3.

For which n does the sum S_n of Problem 1.2 satisfy $S_n = 5n - 6$?

Solution: By part 1.2(a) (proved in Chap. 1 Prob. 2), $S_n = n^2$, so the condition becomes $n^2 - 5n + 6 = 0$. This is the equation of Chap. 1 Prob. 1, whose factorization (1.1) gives $n = 2$ or $n = 3$. Both are positive integers, and indeed $S_2 = 4 = 5 \cdot 2 - 6$ and $S_3 = 9 = 5 \cdot 3 - 6$.

Answer to Problem 3.

$n = 2$ or $n = 3$.

Problem 4.

Show that the sum of the first n even numbers, $2+4+\cdots+2n$, is $n(n+1)$. (Hint: compare it with S_n, or use induction as in (1.2).)